

Problem Set 3: Classical Mechanics 1 (Aug 2026)

3.5 ★★ Many applications of conservation of momentum involve conservation of energy as well, and we haven't yet begun our discussion of energy. Nevertheless, you know enough about energy from your introductory physics course to handle some problems of this type. Here is one elegant example: An *elastic collision* between two bodies is defined as a collision in which the total kinetic energy of the two bodies after the collision is the same as that before. (A familiar example is the collision

between two billiard balls, which generally lose extremely little of their total kinetic energy.) Consider an elastic collision between two equal mass bodies, one of which is initially at rest. Let their velocities be $\mathbf{v}_1$ and $\mathbf{v}_2 = 0$ before the collision, and $\mathbf{v}'_1$ and $\mathbf{v}'_2$ after. Write down the vector equation representing conservation of momentum and the scalar equation which expresses that the collision is elastic. Use these to prove that the angle between $\mathbf{v}'_1$ and $\mathbf{v}'_2$ is 90° . This result was important in the history of atomic and nuclear physics: That two bodies emerged from a collision traveling on perpendicular paths was strongly suggestive that they had equal mass and had undergone an elastic collision.

3.8 ★ A rocket (initial mass m_o) needs to use its engines to hover stationary, just above the ground. **(a)** If it can afford to burn no more than a mass λm_o of its fuel, for how long can it hover? [*Hint:* Write down the condition that the thrust just balance the force of gravity. You can integrate the resulting equation by separating the variables t and m . Take v_{ex} to be constant.] **(b)** If $v_{\text{ex}} \approx 3000$ m/s and $\lambda \approx 10\%$, for how long could the rocket hover just above the earth's surface?

3.11 ★★ **(a)** Consider a rocket traveling in a straight line subject to an external force F^{ext} acting along the same line. Show that the equation of motion is

$$m\dot{v} = -\dot{m}v_{\text{ex}} + F^{\text{ext}}. \quad (3.29)$$

[Review the derivation of Equation (3.6) but keep the external force term.] **(b)** Specialize to the case of a rocket taking off vertically (from rest) in a gravitational field g , so the equation of motion becomes

$$m\dot{v} = -\dot{m}v_{\text{ex}} - mg. \quad (3.30)$$

Assume that the rocket ejects mass at a constant rate, $\dot{m} = -k$ (where k is a positive constant), so that $m = m_o - kt$. Solve equation (3.30) for v as a function of t , using separation of variables (that is, rewriting the equation so that all terms involving v are on the left and all terms involving t on the right). **(c)** Using the rough data from Problem 3.7, find the space shuttle's speed two minutes into flight, assuming (what is nearly true) that it travels vertically up during this period and that g doesn't change appreciably. Compare with the corresponding result if there were no gravity. **(d)** Describe what would

Problem Set 3: Classical Mechanics 1 (Aug 2026)

happen to a rocket that was designed so that the first term on the right of Equation (3.30) was smaller than the initial value of the second.

3.13 ** If you have not already done it, do Problem 3.11(b) and find the speed $v(t)$ of a rocket accelerating vertically from rest in a gravitational field g . Now integrate $v(t)$ and show that the rocket's height as a function of t is

$$y(t) = v_{\text{ex}}t - \frac{1}{2}gt^2 - \frac{mv_{\text{ex}}}{k} \ln\left(\frac{m_0}{m}\right).$$

Using the numbers given in Problem 3.7, estimate the space shuttle's height after two minutes.

3.20 ** Consider a system comprising two extended bodies, which have masses M_1 and M_2 and centers of mass at $\mathbf{R}_1$ and $\mathbf{R}_2$. Prove that the CM of the whole system is at

$$\mathbf{R} = \frac{M_1\mathbf{R}_1 + M_2\mathbf{R}_2}{M_1 + M_2}.$$

This beautiful result means that in finding the CM of a complicated system, you can treat its component parts just like point masses positioned at their separate centers of mass — even when the component parts are themselves extended bodies.

3.21 ** A uniform thin sheet of metal is cut in the shape of a semicircle of radius R and lies in the xy plane with its center at the origin and diameter lying along the x axis. Find the position of the CM using polar coordinates. [In this case the sum (3.9) that defines the CM position becomes a two-dimensional integral of the form $\int \mathbf{r} \sigma dA$ where σ denotes the surface mass density (mass/area) of the sheet and dA is the element of area $dA = r dr d\phi$.]

3.35 ** Consider a uniform solid disk of mass M and radius R , rolling without slipping down an incline which is at angle γ to the horizontal. The instantaneous point of contact between the disk and the incline is called P . (a) Draw a free-body diagram, showing all forces on the disk. (b) Find the linear acceleration $\dot{v}$ of the disk by applying the result $\dot{\mathbf{L}} = \mathbf{\Gamma}^{\text{ext}}$ for rotation about P . (Remember that $L = I\omega$ and the moment of inertia for rotation about a point on the circumference is $\frac{3}{2}MR^2$. The condition that the disk not slip is that $v = R\omega$ and hence $\dot{v} = R\dot{\omega}$.) (c) Derive the same result by applying $\dot{\mathbf{L}} = \mathbf{\Gamma}^{\text{ext}}$

to the rotation about the CM. (In this case you will find there is an extra unknown, the force of friction. You can eliminate this by applying Newton's second law to the motion of the CM. The moment of inertia for rotation about the CM is $\frac{1}{2}MR^2$.)